

Linear Regression Model for Council District I (2023 - 2025 Sales), No Condition

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Model I

Histogram
Dependent Variable: LN_SPPSF

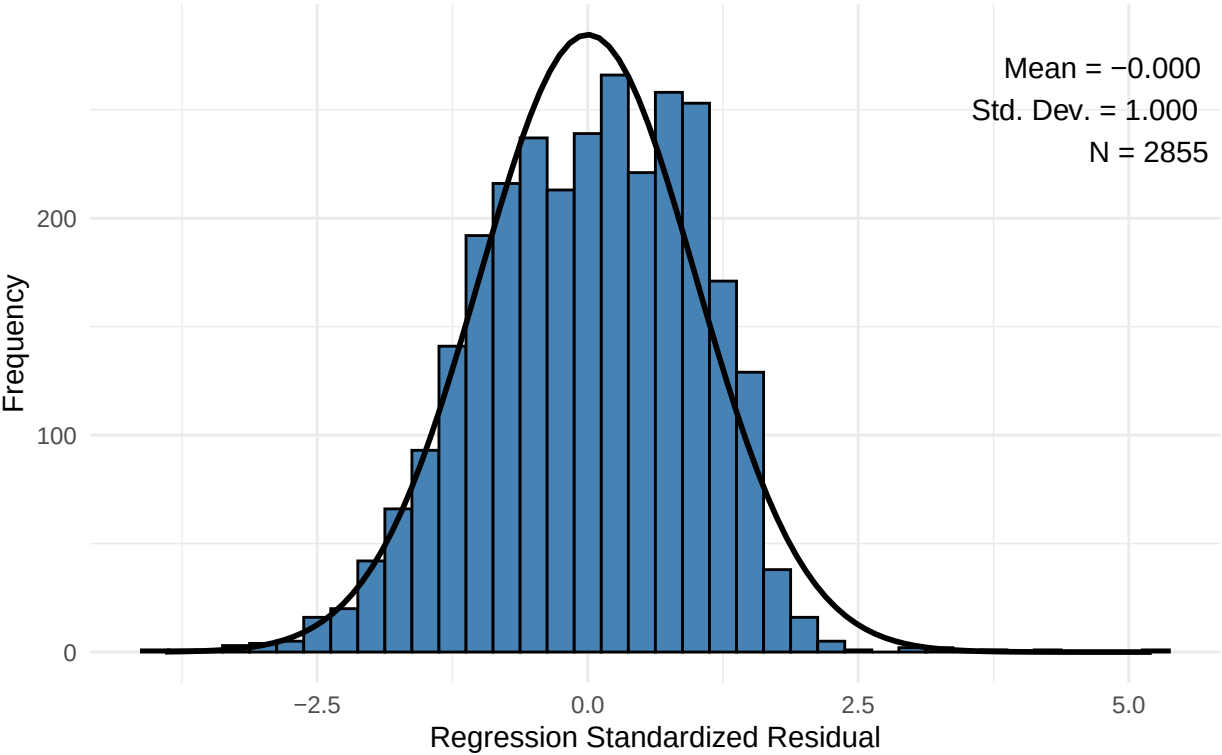


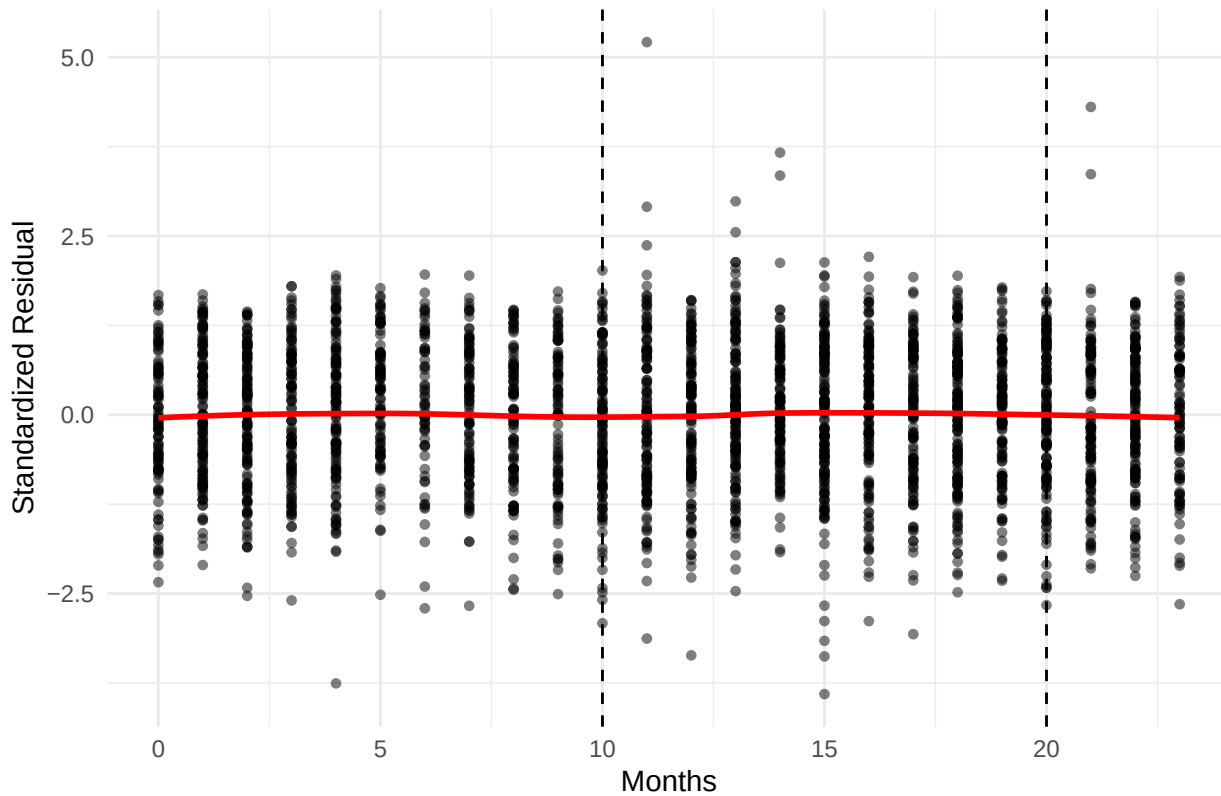
Table 1: District 1 LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2855	2843	12	0.4646	0.0999	0.0965	0.4636	0.3813	28.6994	0	3739.16	3816.599

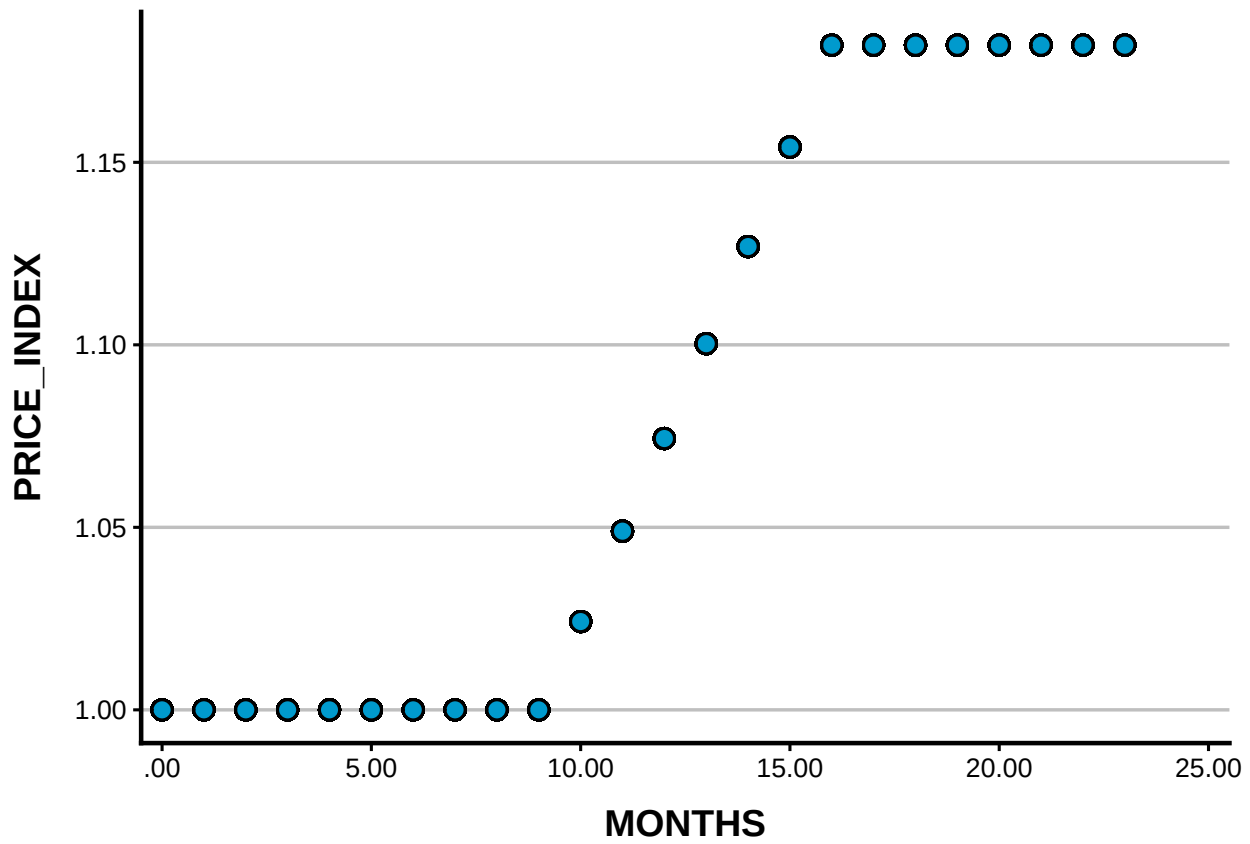
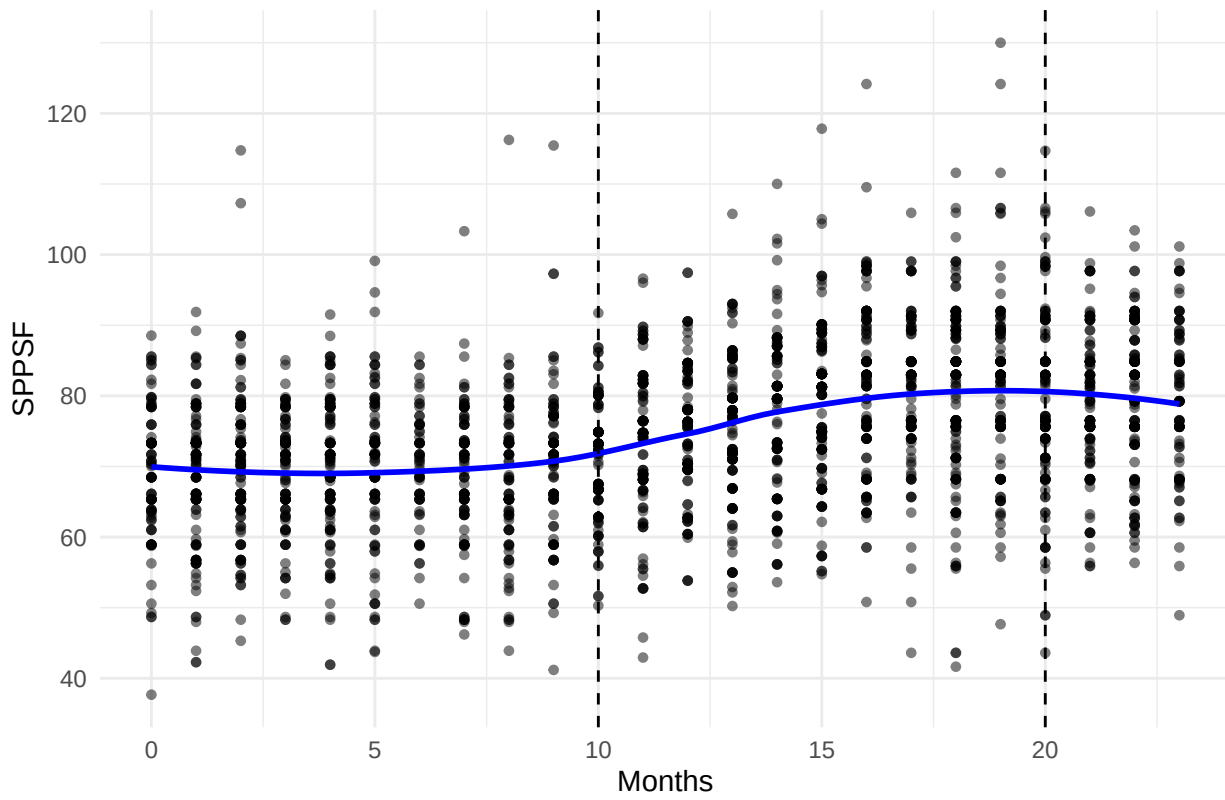
Table 2: District 1 LN_SPPSF model coefficients (95% CI)

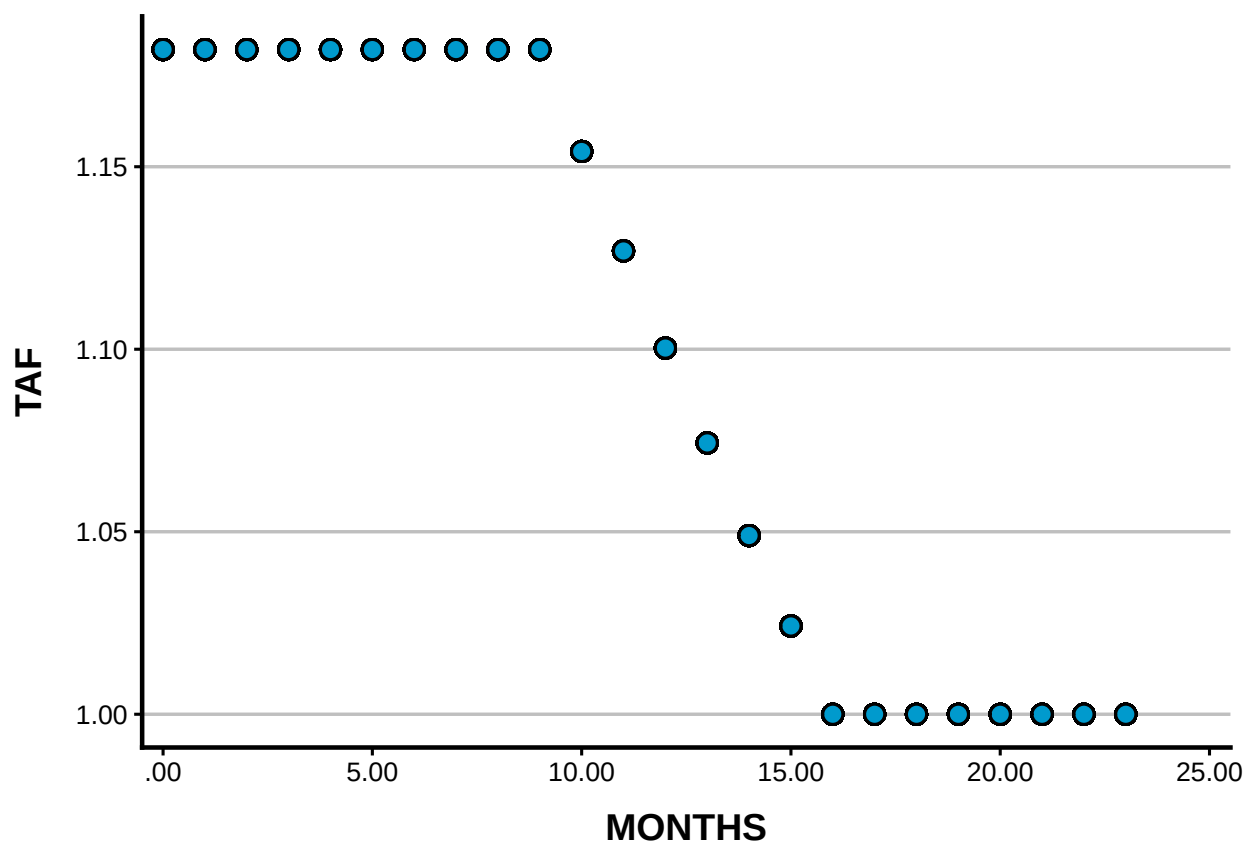
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	4.0302	132.1854	0.0305	0.0000	3.9704	4.0900
single_family1story	0.0460	1.8735	0.0246	0.0611	-0.0021	0.0942
largest	0.0797	2.2738	0.0351	0.0231	0.0110	0.1484
qabove_avg	0.3694	2.4565	0.1504	0.0141	0.0745	0.6643
yb_pre_1929	-0.1528	-5.4579	0.0280	0.0000	-0.2077	-0.0979
yb_1946_to_1960	0.1034	5.1403	0.0201	0.0000	0.0640	0.1429
crawl	-0.2943	-4.9691	0.0592	0.0000	-0.4104	-0.1782
slab	-0.2054	-3.2560	0.0631	0.0011	-0.3290	-0.0817
ln_garagespaces	0.1662	7.3128	0.0227	0.0000	0.1216	0.2108
fireplace	0.0810	4.0774	0.0199	0.0000	0.0420	0.1199
central_air	0.0734	3.0929	0.0237	0.0020	0.0269	0.1199
months_10to16	0.0209	7.3161	0.0029	0.0000	0.0153	0.0265

Residuals for LN_SPPSF Model



Months vs Predicted SPPSF – LN_SPPSF Model





Model II

Histogram
Dependent Variable: LN_PRICE (No ECFs)

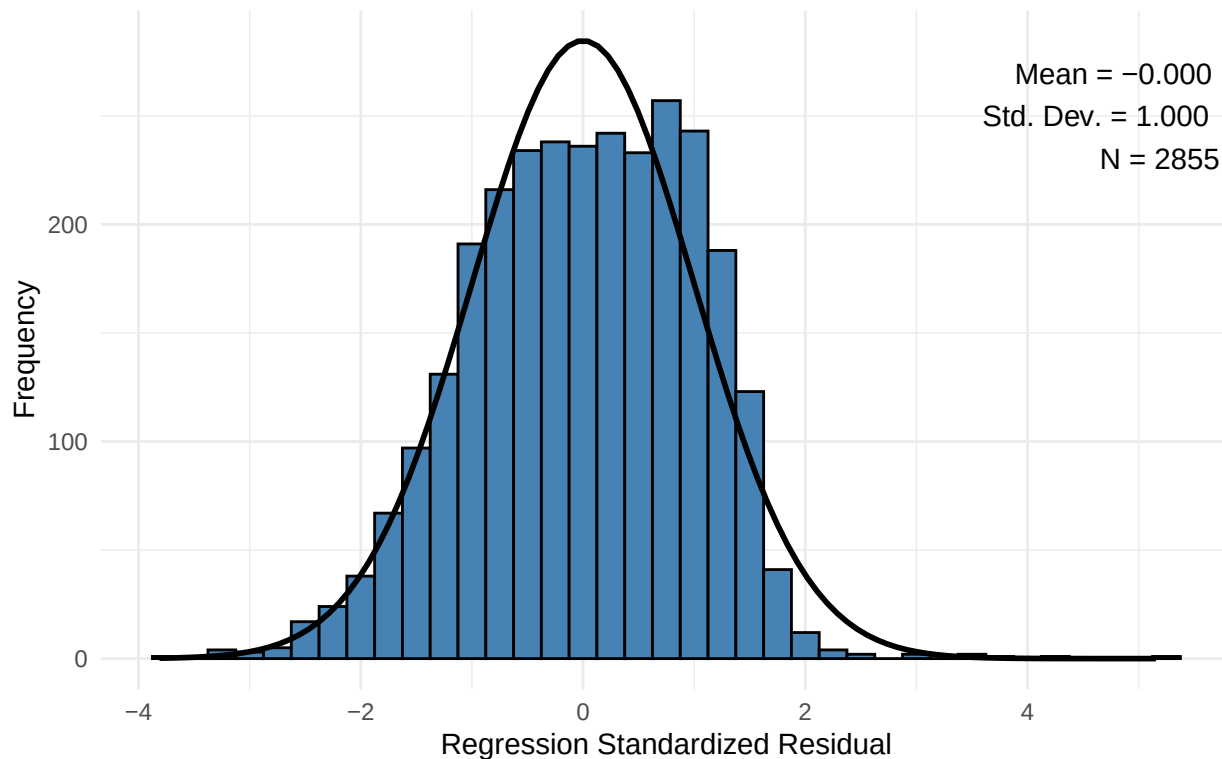


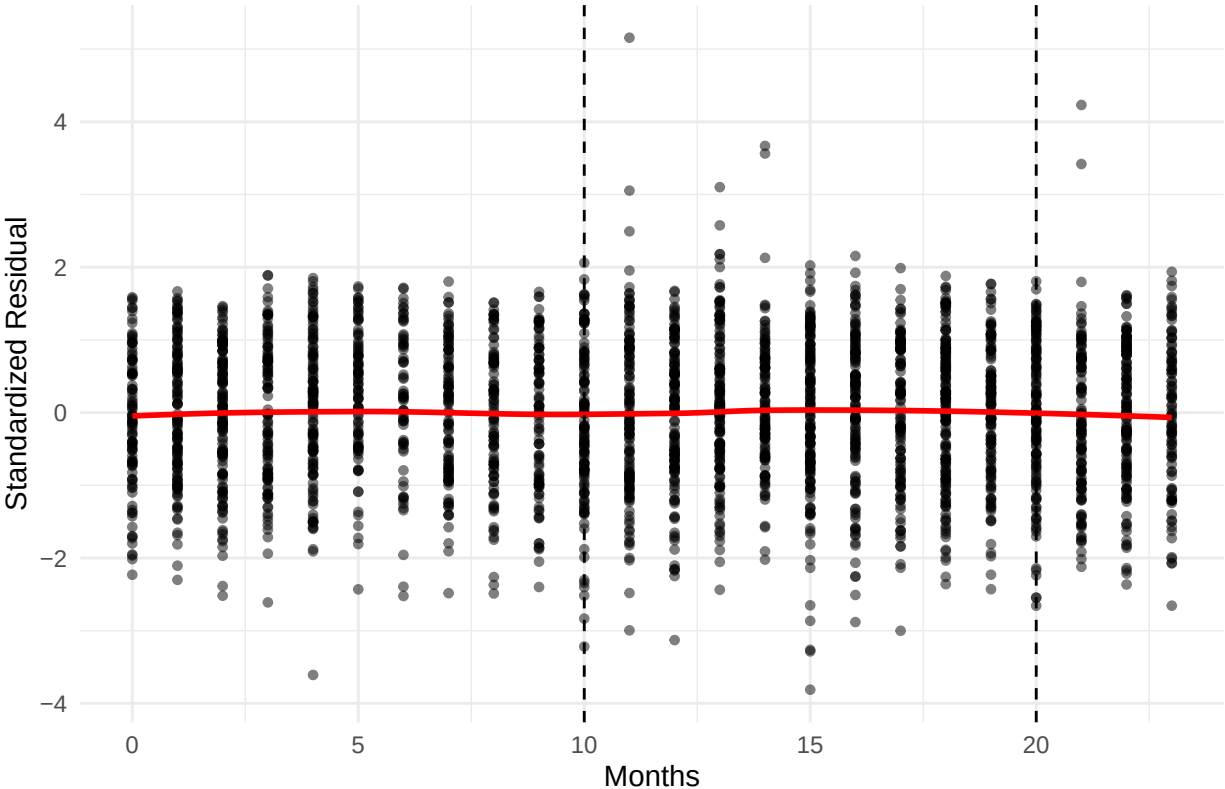
Table 3: District 1 LN_PRICE (No ECFs) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2855	2839	16	0.4614	0.3272	0.3236	0.4601	0.3786	92.0365	0	3703.833	3805.099

Table 4: District 1 LN_PRICE (No ECFs) model coefficients (95% CI)

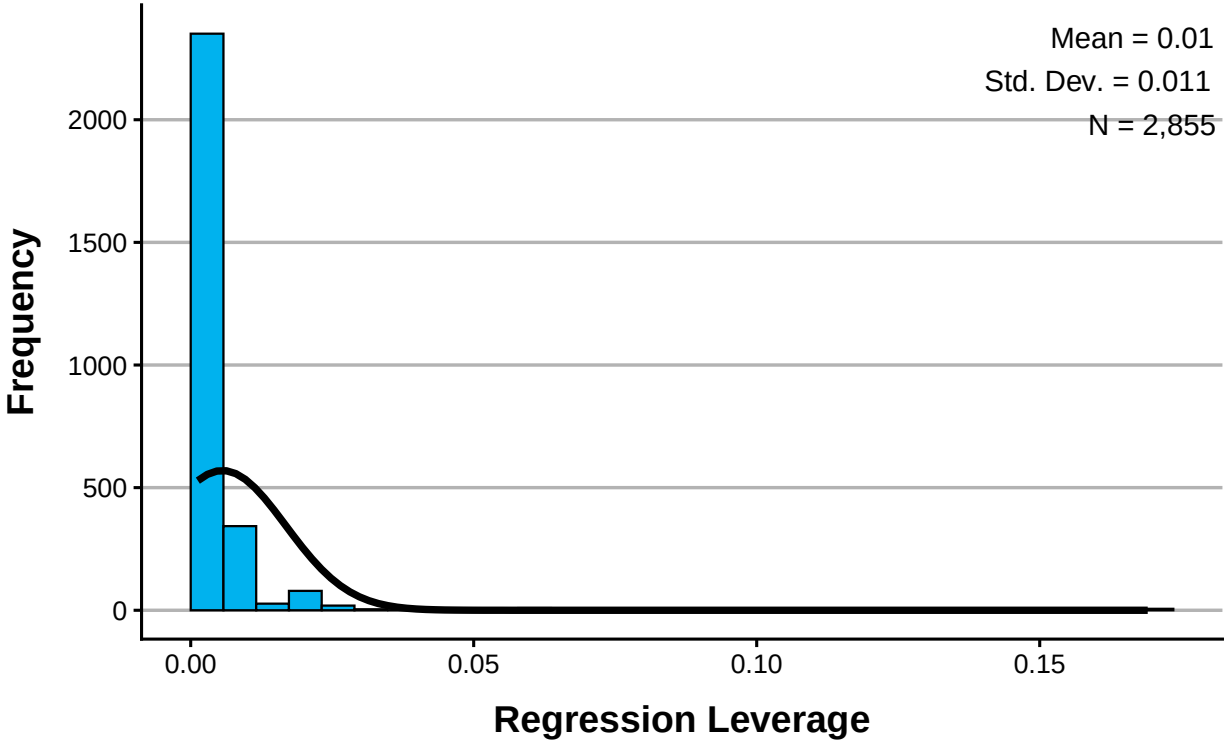
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.0204	416.1501	0.0265	0.0000	10.9684	11.0723
ln_landratio	0.2282	6.0472	0.0377	0.0000	0.1542	0.3022
ln_base_bldgsize_ratio	0.8056	14.4540	0.0557	0.0000	0.6963	0.9149
half_duplex1story	-0.2669	-2.2804	0.1170	0.0227	-0.4964	-0.0374
half_duplex2story	-0.3728	-1.9720	0.1890	0.0487	-0.7435	-0.0021
largest	0.1011	2.4775	0.0408	0.0133	0.0211	0.1812
qabove_avg	0.3246	2.1620	0.1501	0.0307	0.0302	0.6190
qbelow_avg	-0.0361	-1.6604	0.0218	0.0969	-0.0788	0.0065
yb_pre_1929	-0.1471	-5.1744	0.0284	0.0000	-0.2029	-0.0914
yb_1946_to_1960	0.0999	4.9631	0.0201	0.0000	0.0604	0.1393
crawl	-0.2975	-5.0244	0.0592	0.0000	-0.4136	-0.1814
slab	-0.2427	-3.8523	0.0630	0.0001	-0.3662	-0.1192
ln_garagespaces	0.1458	6.3002	0.0231	0.0000	0.1004	0.1911
fireplace	0.1102	4.8983	0.0225	0.0000	0.0661	0.1543
central_air	0.0800	3.3678	0.0238	0.0008	0.0334	0.1266
months_10to16	0.0223	8.0023	0.0028	0.0000	0.0169	0.0278

Residuals for District 1 LN_PRICE (No ECFs) Model

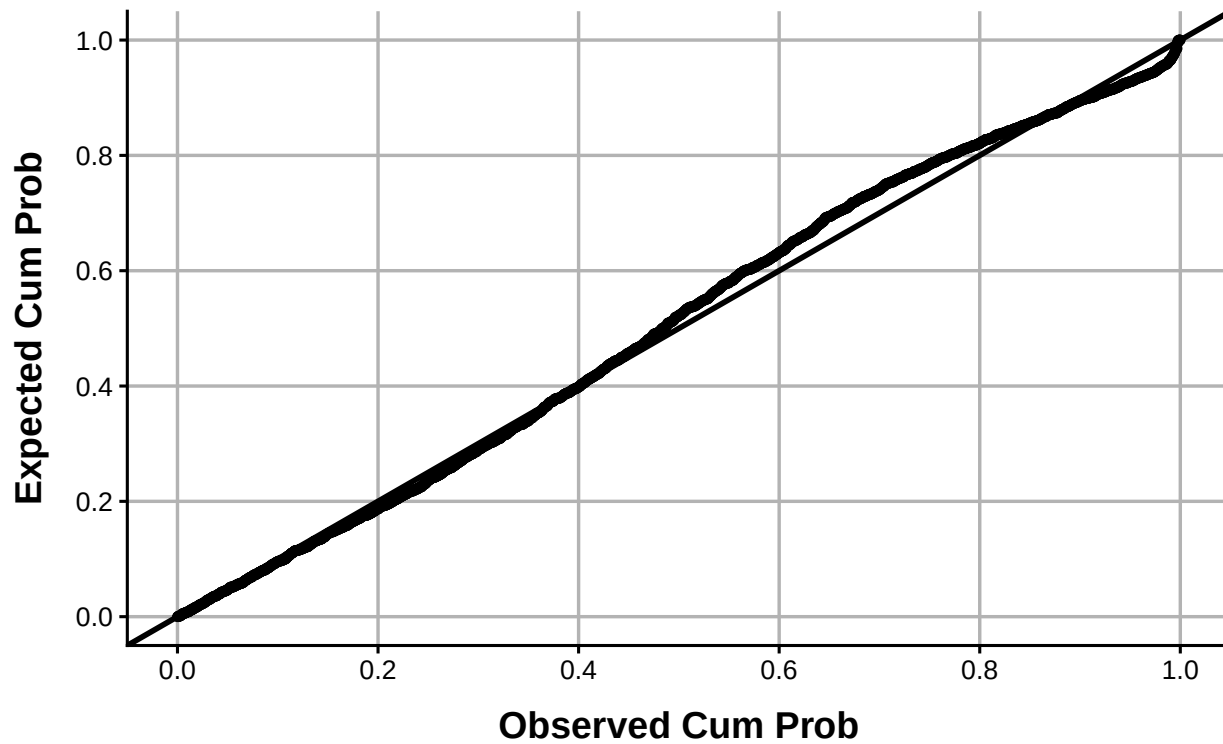


Histogram

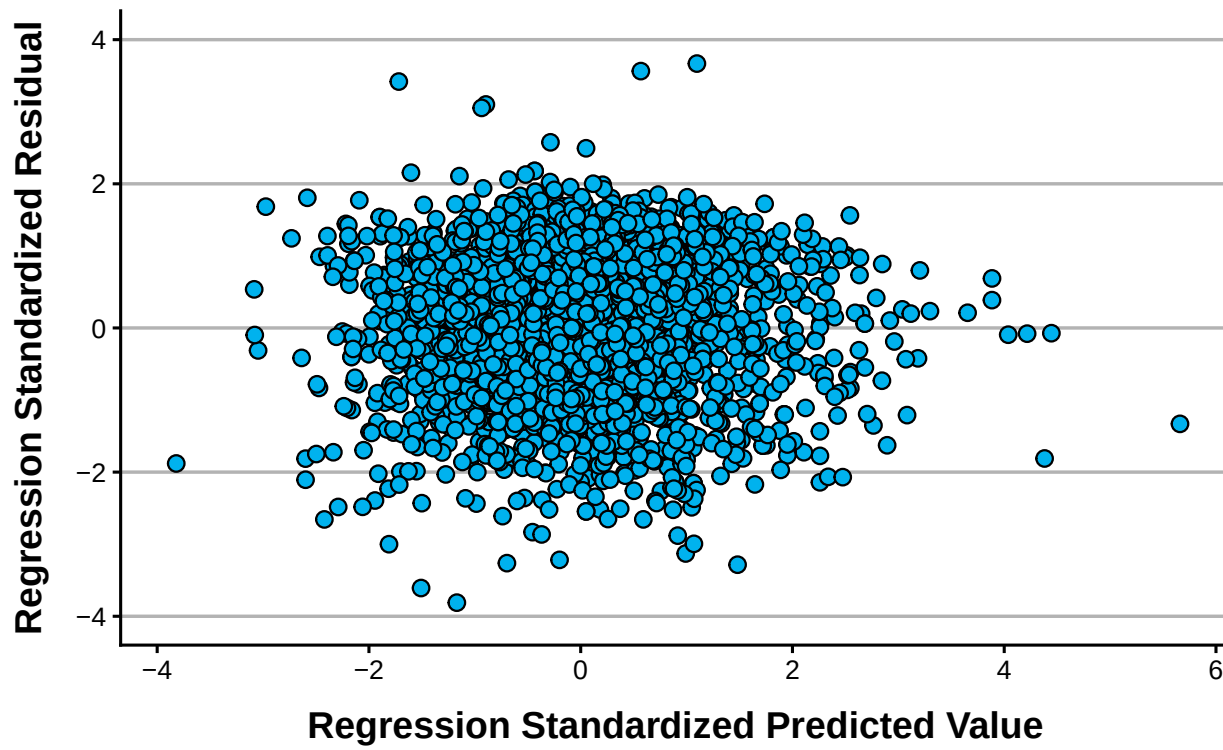
Dependent Variable: LN_PRICE



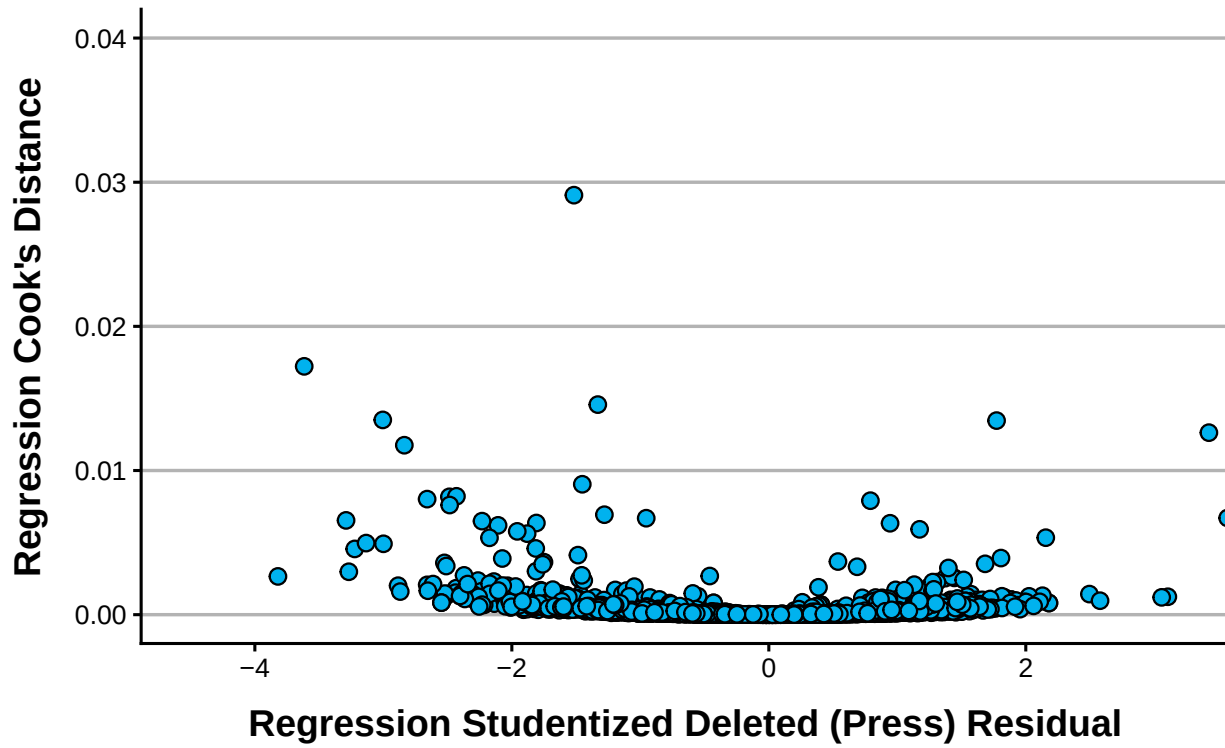
Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



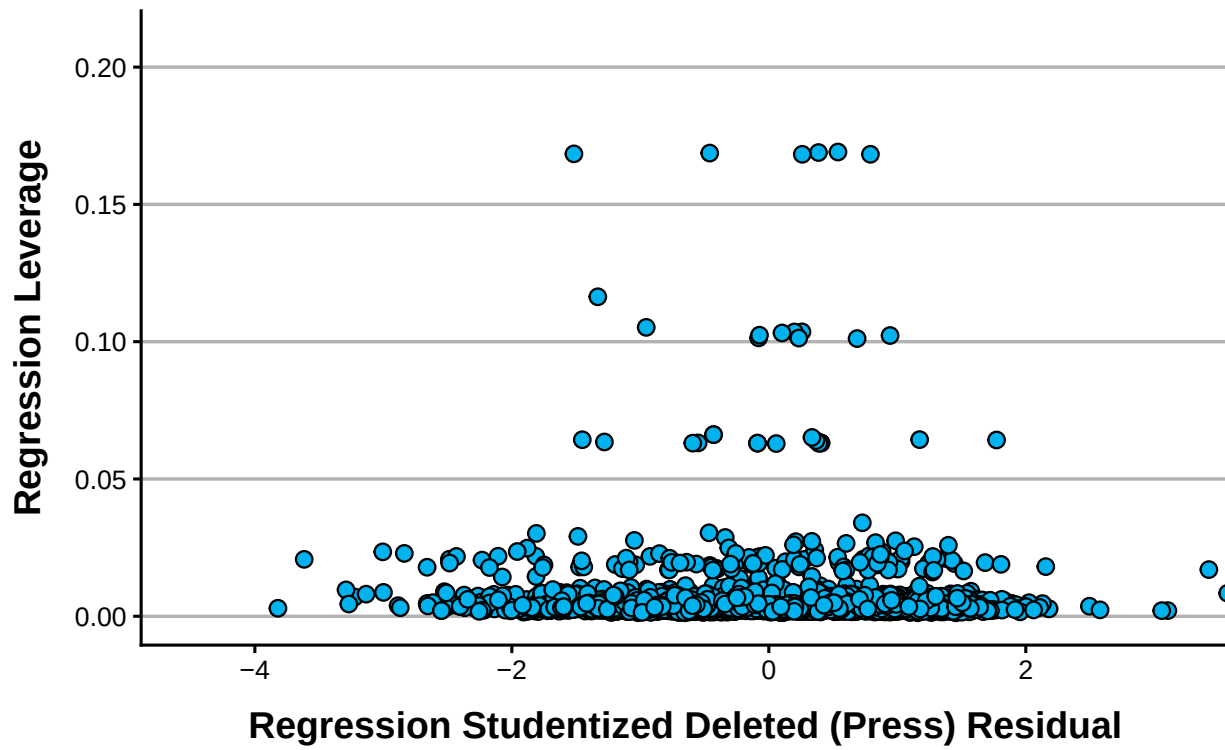
Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Model III

Initial Input

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	0.9857457	1.092391	0.9215637	1.185367	37.45028	-0.6307498

Table 6: District 1 LN_PRICE model fit statistics

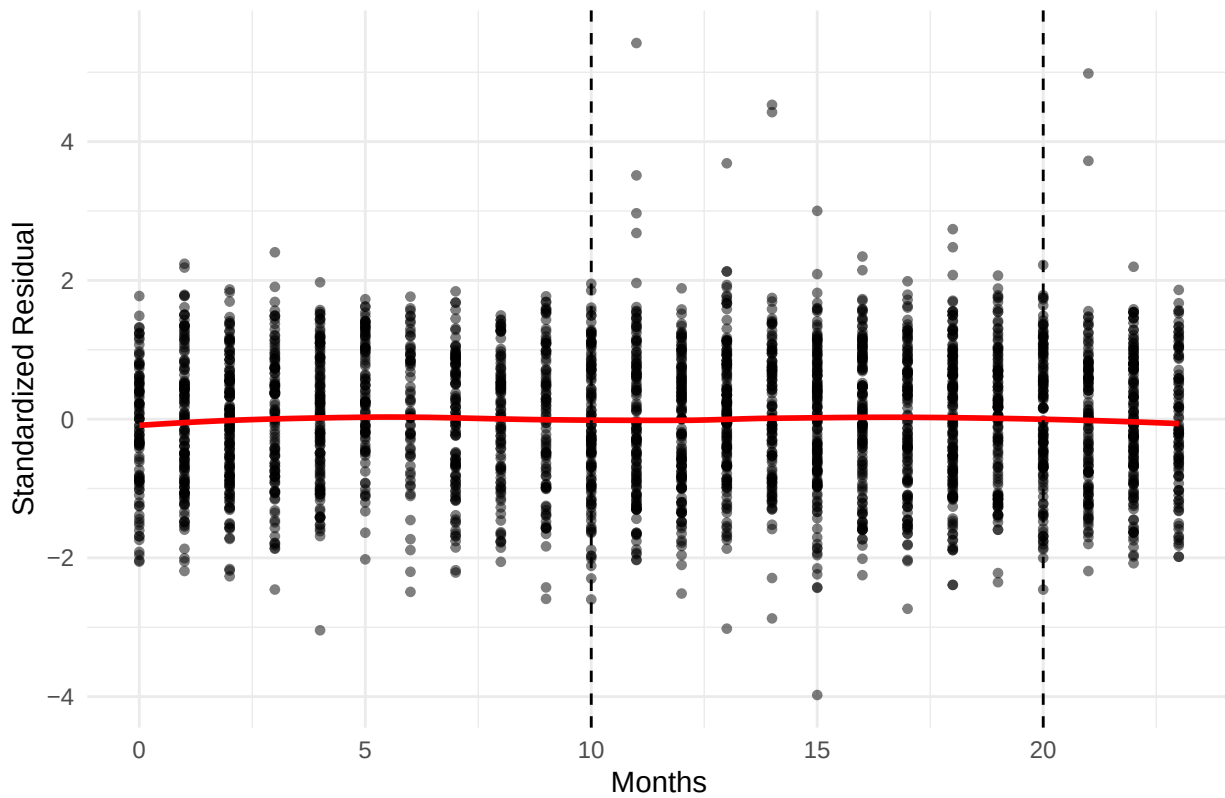
n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2855	2817	38	0.4238	0.4368	0.4294	0.421	0.3478	59.0372	0	3240.315	3472.631

Table 7: District 1 LN_PRICE model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.2076	395.7671	0.0283	0.0000	11.1520	11.2631
ln_landratio	0.0795	2.1009	0.0378	0.0357	0.0053	0.1537
ln_base_bldgsize_ratio	0.7548	16.0678	0.0470	0.0000	0.6627	0.8469
half_duplex1story	-0.3598	-3.3112	0.1086	0.0009	-0.5728	-0.1467
half_duplex2story	-0.5195	-2.9805	0.1743	0.0029	-0.8612	-0.1777
qbelow_avg	-0.0488	-2.2882	0.0213	0.0222	-0.0906	-0.0070
yb_pre_1929	-0.0713	-2.6253	0.0272	0.0087	-0.1246	-0.0181
yb_1946_to_1960	0.0720	3.7733	0.0191	0.0002	0.0346	0.1094
crawl	-0.1820	-3.1283	0.0582	0.0018	-0.2960	-0.0679
slab	-0.1672	-2.7494	0.0608	0.0060	-0.2864	-0.0479
ln_garagespaces	0.1067	4.9565	0.0215	0.0000	0.0645	0.1490
fireplace	0.0582	2.7109	0.0215	0.0068	0.0161	0.1002
central_air	0.0478	2.1759	0.0220	0.0296	0.0047	0.0909
ecf_1R102	-0.2064	-3.0027	0.0687	0.0027	-0.3411	-0.0716
ecf_1R103	0.1377	2.6012	0.0530	0.0093	0.0339	0.2416
ecf_1R105	-0.2053	-5.4674	0.0376	0.0000	-0.2790	-0.1317
ecf_1R106	-0.1384	-2.5017	0.0553	0.0124	-0.2469	-0.0299
ecf_1R108	-0.3001	-7.9887	0.0376	0.0000	-0.3737	-0.2264
ecf_1R113	-0.5048	-7.9968	0.0631	0.0000	-0.6286	-0.3810
ecf_1R114	-0.2106	-3.4446	0.0611	0.0006	-0.3304	-0.0907
ecf_1R115	0.1676	2.3546	0.0712	0.0186	0.0280	0.3072
ecf_1R116	0.3216	6.2512	0.0514	0.0000	0.2207	0.4225
ecf_1R118	-0.1423	-3.2500	0.0438	0.0012	-0.2281	-0.0564
ecf_1R119	-0.2662	-5.2699	0.0505	0.0000	-0.3652	-0.1671
ecf_1R122	-0.2908	-6.7859	0.0428	0.0000	-0.3748	-0.2067
ecf_1R123	-0.1372	-2.3788	0.0577	0.0174	-0.2503	-0.0241
ecf_1R124	-0.5089	-6.1425	0.0829	0.0000	-0.6714	-0.3465
ecf_1R126	0.3695	7.3964	0.0500	0.0000	0.2716	0.4675
ecf_1R127	-0.2666	-4.2292	0.0630	0.0000	-0.3902	-0.1430
ecf_1R128	-0.4096	-10.0455	0.0408	0.0000	-0.4896	-0.3297
ecf_1R130	-0.5949	-3.3088	0.1798	0.0009	-0.9474	-0.2423
ecf_1R131	-0.1839	-1.7595	0.1045	0.0786	-0.3888	0.0210
ecf_1R133	-0.2537	-3.0404	0.0834	0.0024	-0.4173	-0.0901
ecf_1R134	0.2912	3.8888	0.0749	0.0001	0.1444	0.4380
ecf_1R135	-0.1129	-2.3698	0.0476	0.0179	-0.2063	-0.0195
ecf_1R136	-0.1558	-3.8103	0.0409	0.0001	-0.2360	-0.0756
ecf_1R137	-0.7166	-10.3848	0.0690	0.0000	-0.8519	-0.5813

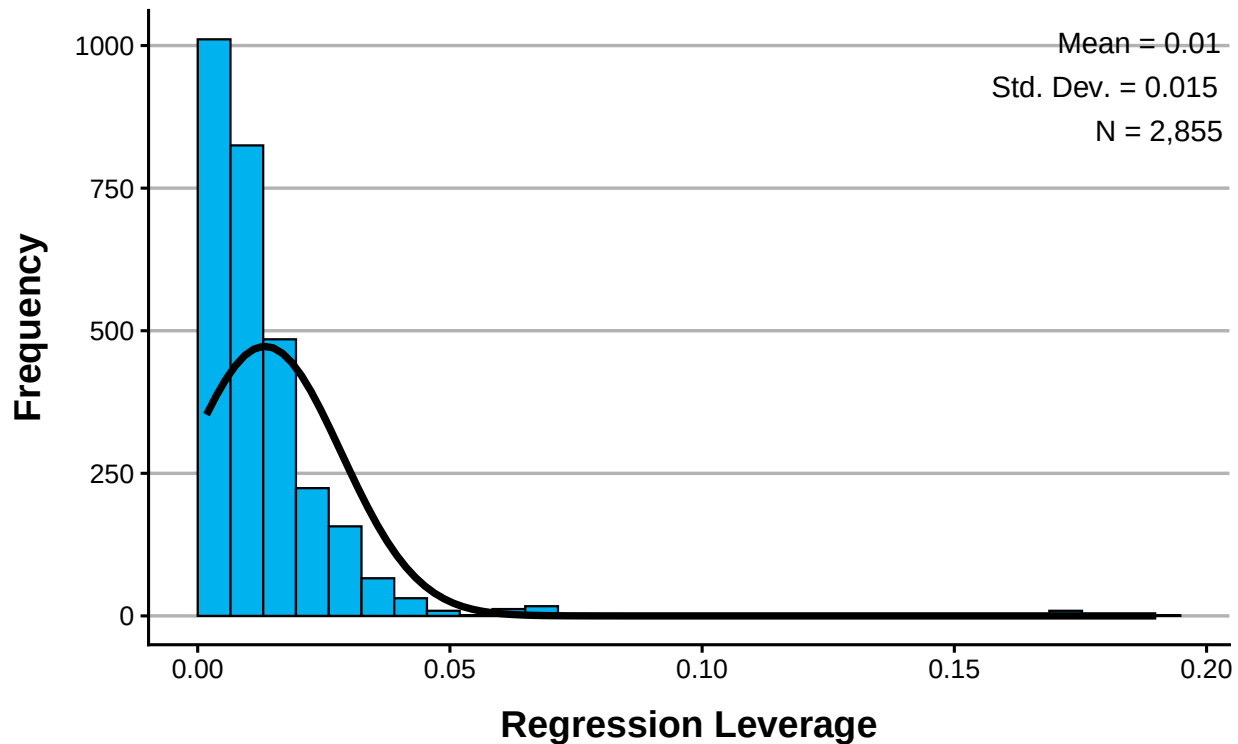
term	estimate	statistic	std_error	p_value	conf_low	conf_high
months_10to16	0.0230	8.9177	0.0026	0.0000	0.0179	0.0280

Residuals for District 1 LN_PRICE Model

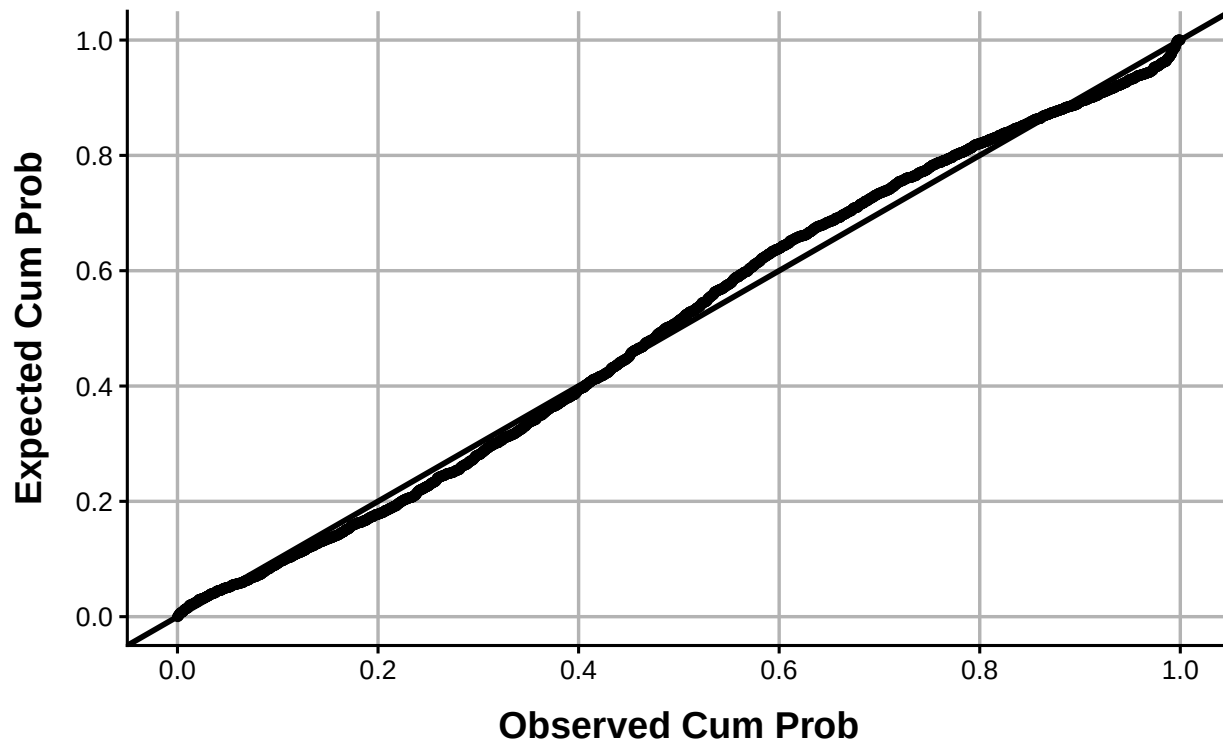


Histogram

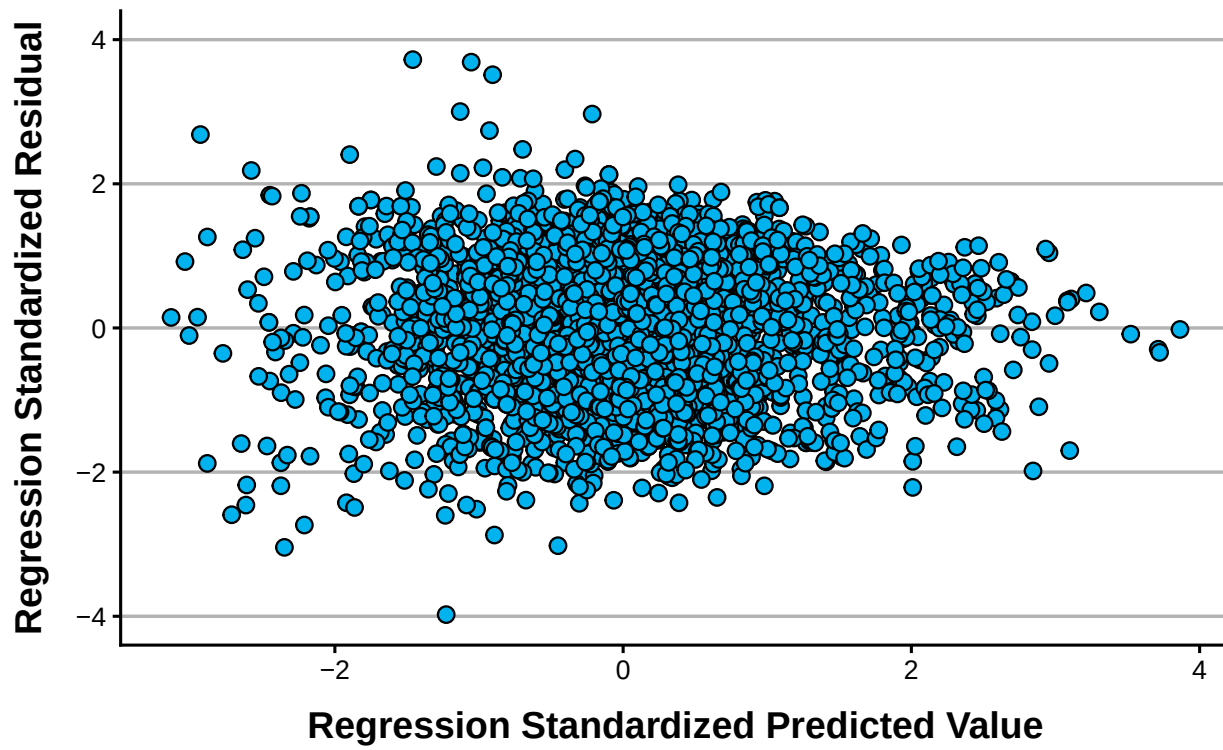
Dependent Variable: LN_PRICE

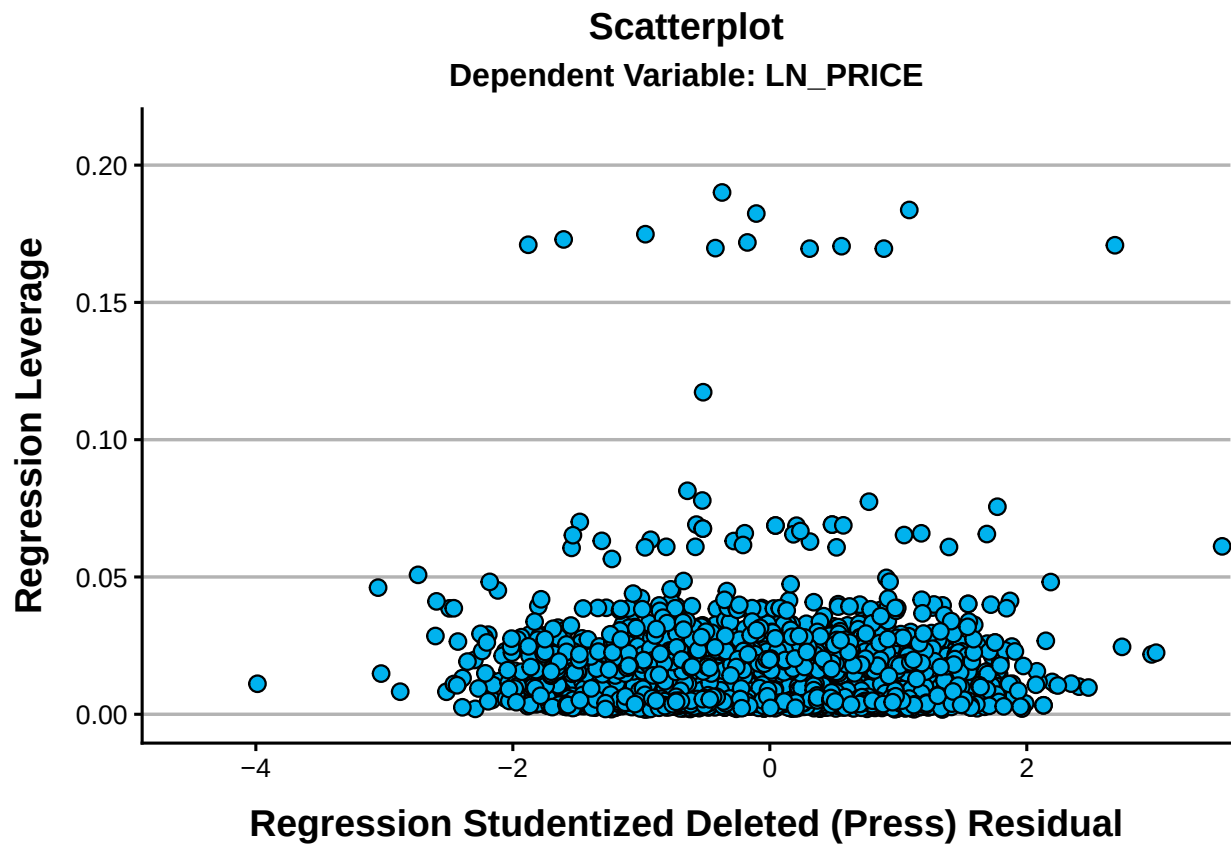
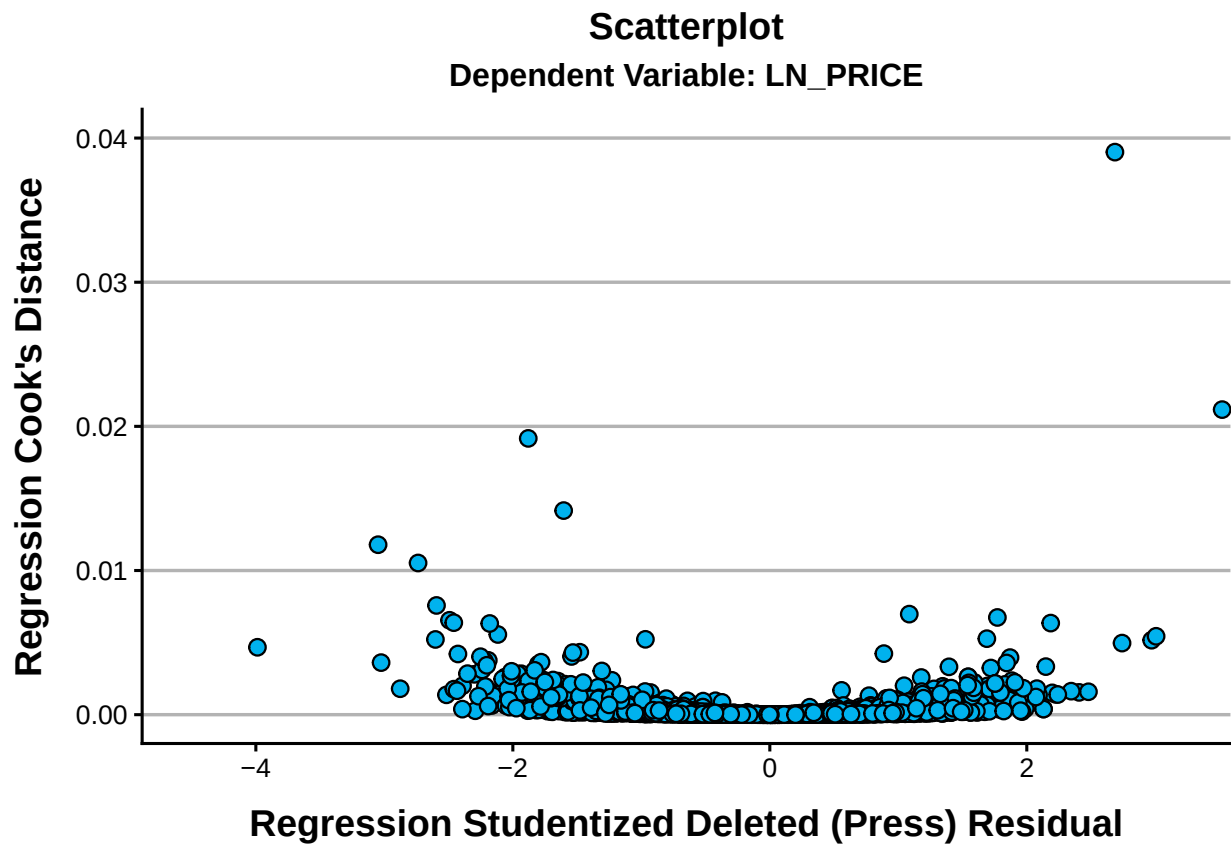


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Filtered Input

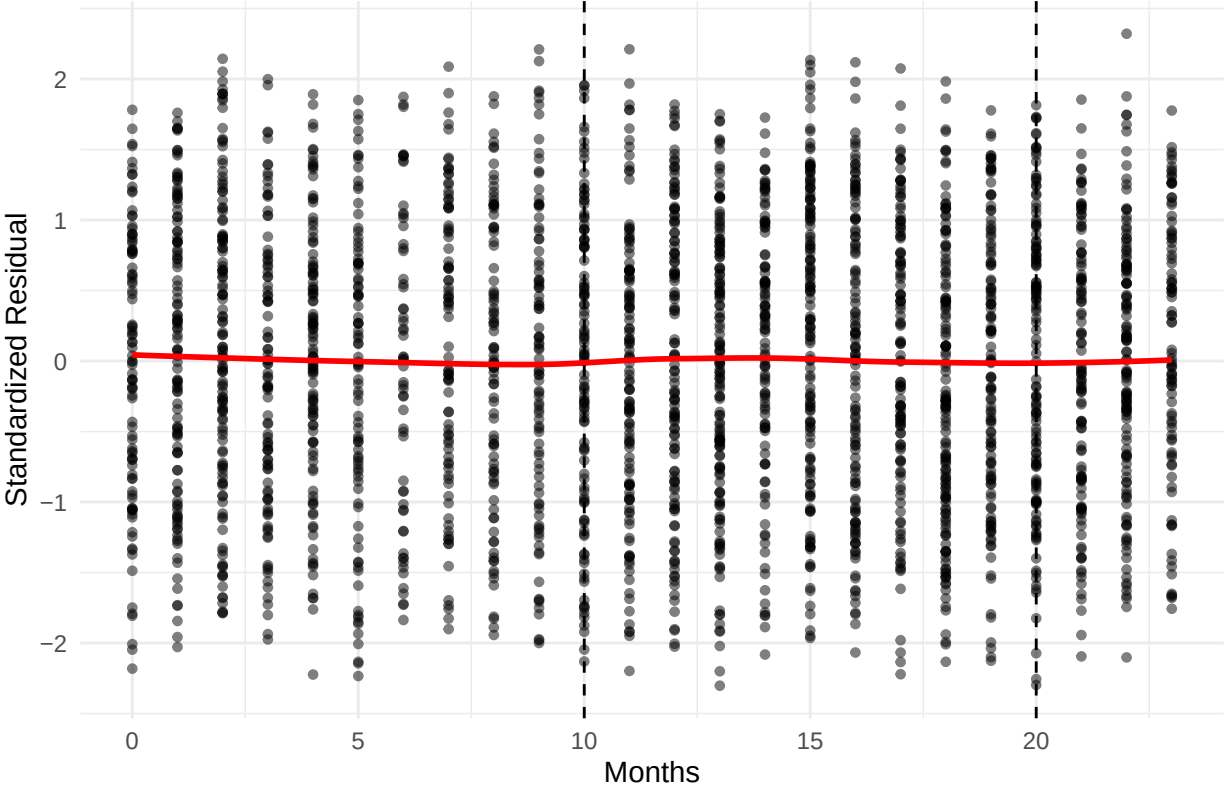
Table 8: District 1 LN_PRICE (Inliers) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2686	2647	39	0.3754	0.4876	0.4803	0.3726	0.3149	66.2931	0	2399.273	2635.105

Table 9: District 1 LN_PRICE (Inliers) model coefficients (95% CI)

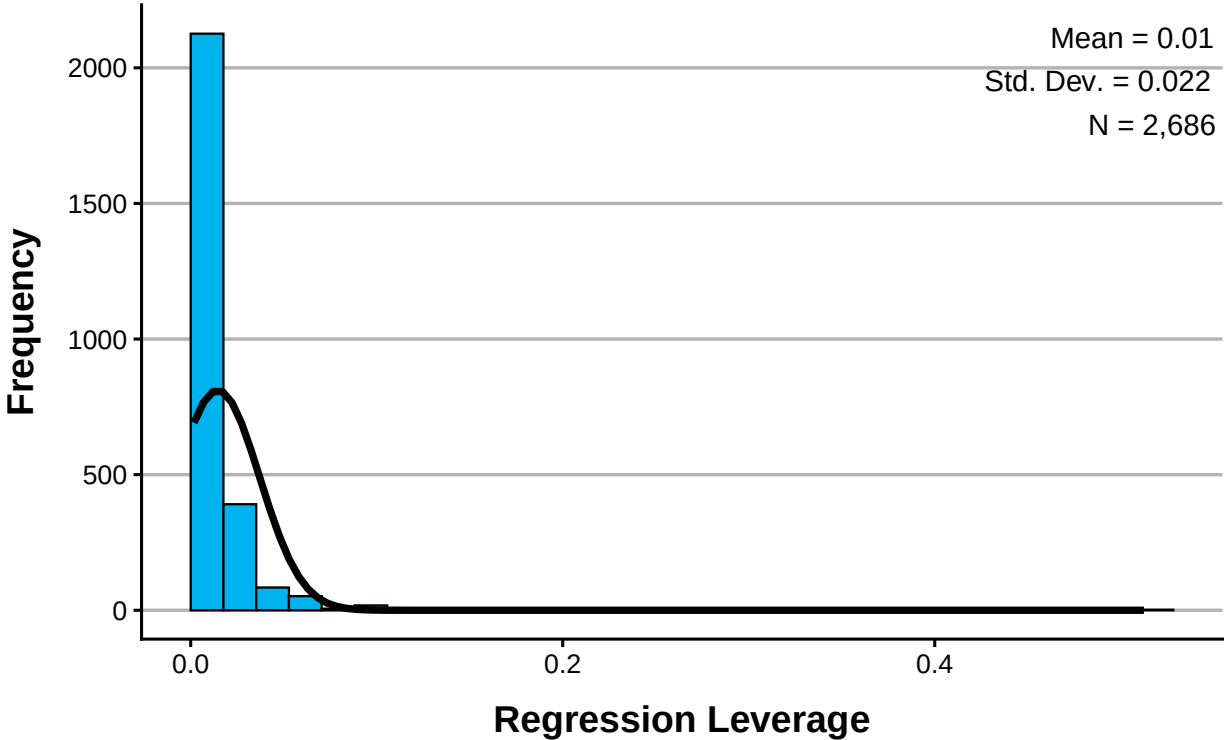
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.2412	435.2788	0.0258	0.0000	11.1905	11.2918
ln_landratio	0.0821	2.2900	0.0359	0.0221	0.0118	0.1525
ln_base_bldgsize_ratio	0.7579	17.4316	0.0435	0.0000	0.6726	0.8432
half_duplex1story	-0.3693	-3.3214	0.1112	0.0009	-0.5873	-0.1513
half_duplex2story	-0.5960	-2.7290	0.2184	0.0064	-1.0242	-0.1678
qbelow_avg	-0.0650	-3.3437	0.0194	0.0008	-0.1031	-0.0269
yb_pre_1929	-0.0631	-2.4942	0.0253	0.0127	-0.1127	-0.0135
yb_1946_to_1960	0.0545	3.1349	0.0174	0.0017	0.0204	0.0886
crawl	-0.1520	-2.5309	0.0600	0.0114	-0.2697	-0.0342
slab	-0.1526	-2.4762	0.0616	0.0133	-0.2735	-0.0318
ln_garagespaces	0.1163	5.8852	0.0198	0.0000	0.0776	0.1551
fireplace	0.0431	2.1990	0.0196	0.0280	0.0047	0.0816
central_air	0.0566	2.8595	0.0198	0.0043	0.0178	0.0955
ecf_1R102	-0.1764	-2.5069	0.0704	0.0122	-0.3144	-0.0384
ecf_1R103	0.1658	3.4535	0.0480	0.0006	0.0717	0.2600
ecf_1R105	-0.2123	-6.3432	0.0335	0.0000	-0.2780	-0.1467
ecf_1R106	-0.1493	-2.9508	0.0506	0.0032	-0.2485	-0.0501
ecf_1R107	-0.1807	-1.9400	0.0932	0.0525	-0.3634	0.0019
ecf_1R108	-0.2901	-8.4511	0.0343	0.0000	-0.3574	-0.2228
ecf_1R113	-0.4927	-7.1453	0.0690	0.0000	-0.6279	-0.3575
ecf_1R114	-0.1050	-1.7482	0.0600	0.0805	-0.2227	0.0128
ecf_1R115	0.2425	3.5450	0.0684	0.0004	0.1084	0.3767
ecf_1R116	0.3002	6.5399	0.0459	0.0000	0.2102	0.3902
ecf_1R118	-0.1185	-2.9863	0.0397	0.0028	-0.1962	-0.0407
ecf_1R119	-0.3596	-7.6102	0.0473	0.0000	-0.4523	-0.2670
ecf_1R122	-0.2800	-7.1873	0.0390	0.0000	-0.3565	-0.2036
ecf_1R123	-0.1068	-1.9744	0.0541	0.0484	-0.2128	-0.0007
ecf_1R124	-0.4814	-5.3664	0.0897	0.0000	-0.6573	-0.3055
ecf_1R126	0.3715	8.2747	0.0449	0.0000	0.2834	0.4595
ecf_1R127	-0.3258	-5.3694	0.0607	0.0000	-0.4447	-0.2068
ecf_1R128	-0.4614	-12.4489	0.0371	0.0000	-0.5341	-0.3887
ecf_1R130	-0.6709	-2.4893	0.2695	0.0129	-1.1995	-0.1424
ecf_1R131	-0.2782	-2.4154	0.1152	0.0158	-0.5041	-0.0524
ecf_1R133	-0.3454	-3.6242	0.0953	0.0003	-0.5322	-0.1585
ecf_1R134	0.2899	4.2838	0.0677	0.0000	0.1572	0.4226
ecf_1R135	-0.1150	-2.7020	0.0426	0.0069	-0.1985	-0.0316
ecf_1R136	-0.1554	-4.2604	0.0365	0.0000	-0.2269	-0.0839
ecf_1R137	-0.6142	-8.9079	0.0689	0.0000	-0.7494	-0.4790
months_10to16	0.0201	8.5254	0.0024	0.0000	0.0155	0.0247

Residuals for LN_PRICE (Inliers) Model

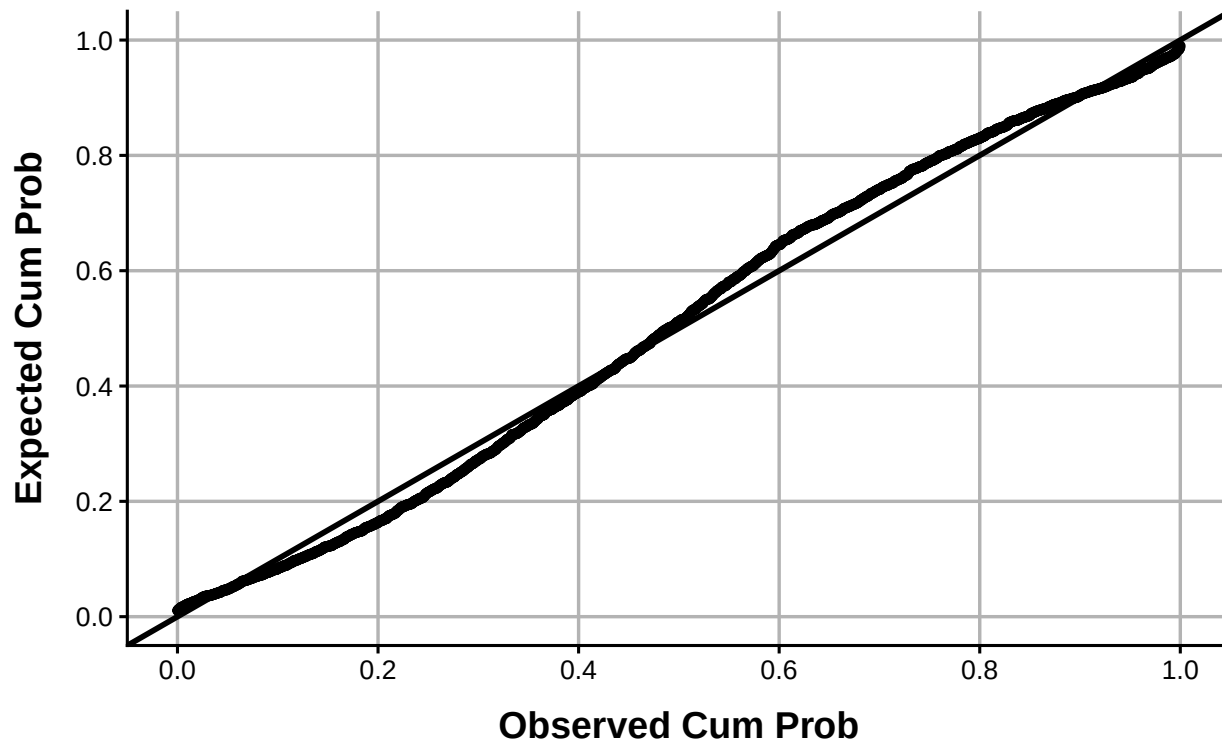


Histogram

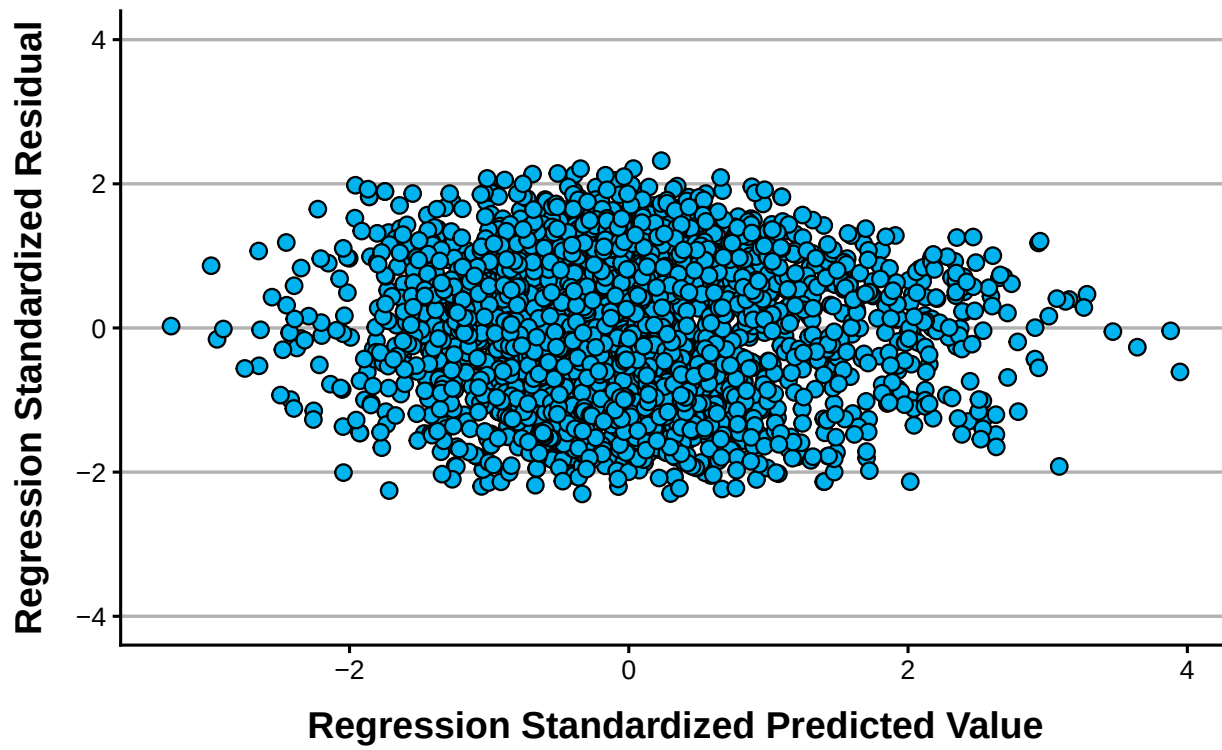
Dependent Variable: LN_PRICE



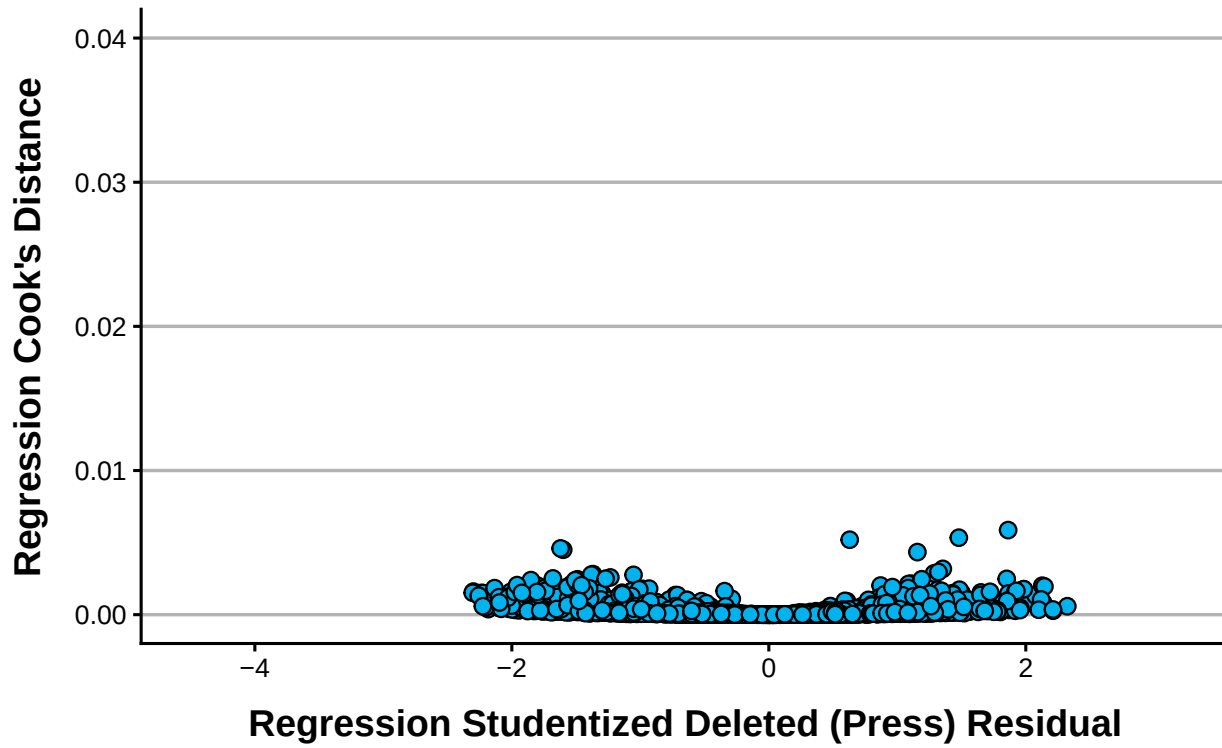
Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



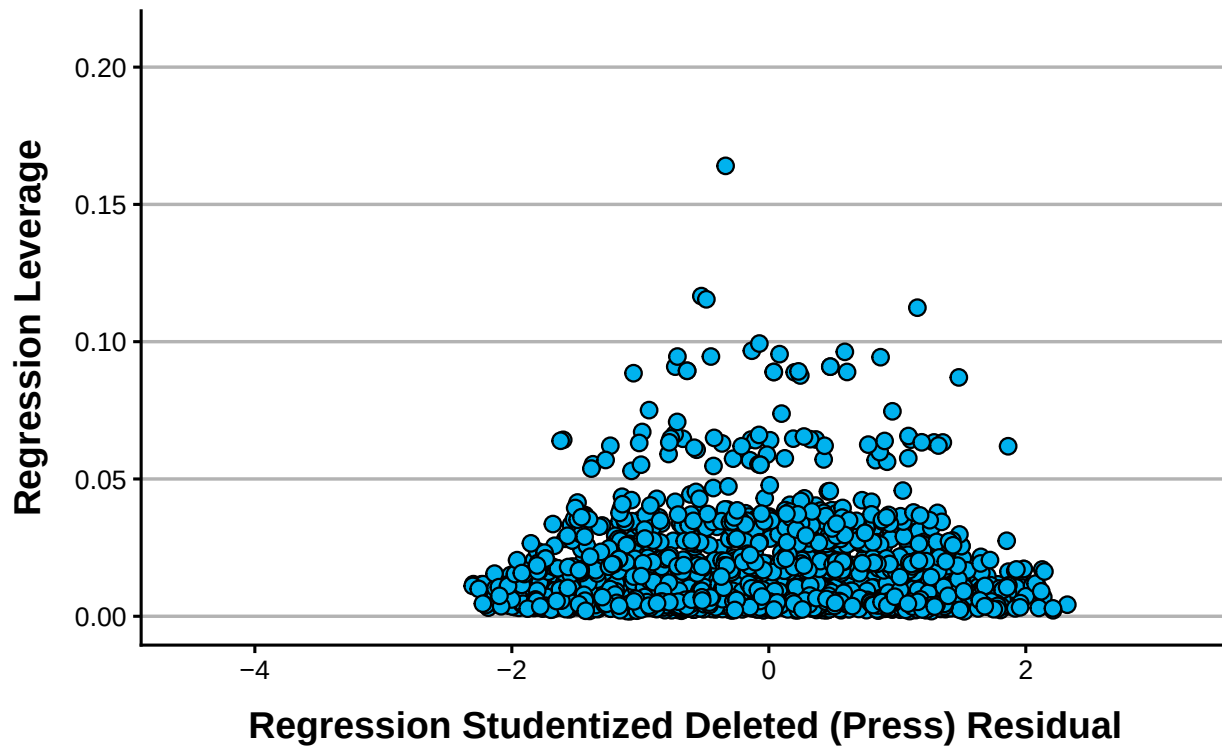
Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Model IV

Table 10: District 1 LN_TASP Model I assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	1.084782	1.199207	1.008391	1.189229	37.58676	-0.708339

Table 11: District 1 LN_TASP Model II assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	1.092324	1.198735	1.007972	1.189254	37.33184	-0.7069979

Table 12: District 1 LN_TASP Model I fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2686	2648	38	0.3754	0.4806	0.4734	0.3727	0.315	66.2333	0	2398.553	2628.49

Table 13: District 1 LN_TASP Model I coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.3906	467.1969	0.0244	0.0000	11.3428	11.4384
ln_landratio	0.0831	2.3191	0.0359	0.0205	0.0128	0.1535
ln_base_bldgsize_ratio	0.7571	17.4148	0.0435	0.0000	0.6719	0.8424
half_duplex1story	-0.3718	-3.3447	0.1112	0.0008	-0.5898	-0.1538
half_duplex2story	-0.5936	-2.7183	0.2184	0.0066	-1.0218	-0.1654
qbelow_avg	-0.0651	-3.3521	0.0194	0.0008	-0.1032	-0.0270
yb_pre_1929	-0.0642	-2.5367	0.0253	0.0112	-0.1137	-0.0146
yb_1946_to_1960	0.0548	3.1510	0.0174	0.0016	0.0207	0.0889
crawl	-0.1505	-2.5073	0.0600	0.0122	-0.2682	-0.0328
slab	-0.1541	-2.5001	0.0616	0.0125	-0.2749	-0.0332
ln_garagespaces	0.1173	5.9412	0.0197	0.0000	0.0786	0.1560
fireplace	0.0431	2.1990	0.0196	0.0280	0.0047	0.0816
central_air	0.0565	2.8526	0.0198	0.0044	0.0177	0.0953
ecf_1R102	-0.1767	-2.5116	0.0704	0.0121	-0.3147	-0.0388
ecf_1R103	0.1637	3.4112	0.0480	0.0007	0.0696	0.2578
ecf_1R105	-0.2133	-6.3735	0.0335	0.0000	-0.2789	-0.1477
ecf_1R106	-0.1482	-2.9301	0.0506	0.0034	-0.2475	-0.0490
ecf_1R107	-0.1809	-1.9416	0.0932	0.0523	-0.3636	0.0018
ecf_1R108	-0.2908	-8.4722	0.0343	0.0000	-0.3581	-0.2235
ecf_1R113	-0.4927	-7.1452	0.0690	0.0000	-0.6279	-0.3575
ecf_1R114	-0.1035	-1.7247	0.0600	0.0847	-0.2213	0.0142
ecf_1R115	0.2409	3.5214	0.0684	0.0004	0.1067	0.3750
ecf_1R116	0.3004	6.5447	0.0459	0.0000	0.2104	0.3904
ecf_1R118	-0.1180	-2.9741	0.0397	0.0030	-0.1958	-0.0402
ecf_1R119	-0.3602	-7.6222	0.0473	0.0000	-0.4529	-0.2675
ecf_1R122	-0.2801	-7.1881	0.0390	0.0000	-0.3565	-0.2037
ecf_1R123	-0.1061	-1.9615	0.0541	0.0499	-0.2121	0.0000
ecf_1R124	-0.4808	-5.3590	0.0897	0.0000	-0.6567	-0.3049
ecf_1R126	0.3714	8.2718	0.0449	0.0000	0.2833	0.4594
ecf_1R127	-0.3271	-5.3917	0.0607	0.0000	-0.4460	-0.2081
ecf_1R128	-0.4592	-12.4063	0.0370	0.0000	-0.5318	-0.3866
ecf_1R130	-0.6618	-2.4564	0.2694	0.0141	-1.1901	-0.1335
ecf_1R131	-0.2790	-2.4215	0.1152	0.0155	-0.5048	-0.0531

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_1R133	-0.3460	-3.6307	0.0953	0.0003	-0.5329	-0.1591
ecf_1R134	0.2917	4.3109	0.0677	0.0000	0.1590	0.4243
ecf_1R135	-0.1154	-2.7097	0.0426	0.0068	-0.1989	-0.0319
ecf_1R136	-0.1566	-4.2949	0.0365	0.0000	-0.2280	-0.0851
ecf_1R137	-0.6173	-8.9603	0.0689	0.0000	-0.7524	-0.4822

Table 14: District 1 LN_TASP Model II fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
2686	2645	41	0.3754	0.481	0.4732	0.3726	0.3152	61.2897	0	2402.585	2650.209

Table 15: District 1 LN_TASP Model II coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	8.7865	40.3879	0.2176	0.0000	8.3600	9.2131
ln_landratio	0.0982	2.7403	0.0358	0.0062	0.0279	0.1685
lnsbldsfint	0.4036	11.7754	0.0343	0.0000	0.3364	0.4708
half_duplex1story	-0.1928	-1.7280	0.1116	0.0841	-0.4115	0.0260
half_duplex2story	-0.4322	-1.9792	0.2184	0.0479	-0.8605	-0.0040
large	0.0488	1.9307	0.0253	0.0536	-0.0008	0.0984
largest	0.0939	2.2028	0.0426	0.0277	0.0103	0.1775
qabove_avg	0.2364	1.9166	0.1233	0.0554	-0.0055	0.4782
qbelow_avg	-0.0619	-3.1478	0.0197	0.0017	-0.1004	-0.0233
yb_pre_1929	-0.0616	-2.4302	0.0253	0.0152	-0.1113	-0.0119
yb_1946_to_1960	0.0560	3.1754	0.0176	0.0015	0.0214	0.0905
crawl	-0.1439	-2.3907	0.0602	0.0169	-0.2619	-0.0259
slab	-0.1433	-2.3162	0.0619	0.0206	-0.2647	-0.0220
garage	0.0961	5.6132	0.0171	0.0000	0.0625	0.1296
fireplace	0.0402	2.0366	0.0197	0.0418	0.0015	0.0789
central_air	0.0596	3.0075	0.0198	0.0027	0.0207	0.0984
ecf_1R102	-0.1837	-2.6098	0.0704	0.0091	-0.3217	-0.0457
ecf_1R103	0.1663	3.4643	0.0480	0.0005	0.0722	0.2605
ecf_1R105	-0.2096	-6.2416	0.0336	0.0000	-0.2755	-0.1438
ecf_1R106	-0.1517	-2.9977	0.0506	0.0027	-0.2510	-0.0525
ecf_1R107	-0.1811	-1.9428	0.0932	0.0521	-0.3640	0.0017
ecf_1R108	-0.2854	-8.2970	0.0344	0.0000	-0.3528	-0.2179
ecf_1R113	-0.4841	-7.0162	0.0690	0.0000	-0.6193	-0.3488
ecf_1R114	-0.1037	-1.7247	0.0601	0.0847	-0.2217	0.0142
ecf_1R115	0.2265	3.2956	0.0687	0.0010	0.0917	0.3612
ecf_1R116	0.3056	6.5042	0.0470	0.0000	0.2135	0.3978
ecf_1R118	-0.1211	-3.0497	0.0397	0.0023	-0.1990	-0.0432
ecf_1R119	-0.3650	-7.7184	0.0473	0.0000	-0.4577	-0.2722
ecf_1R122	-0.2771	-7.1095	0.0390	0.0000	-0.3535	-0.2007
ecf_1R123	-0.1093	-2.0188	0.0542	0.0436	-0.2155	-0.0031
ecf_1R124	-0.4772	-5.3120	0.0898	0.0000	-0.6534	-0.3011
ecf_1R126	0.3669	8.1283	0.0451	0.0000	0.2784	0.4554
ecf_1R127	-0.3277	-5.3856	0.0609	0.0000	-0.4470	-0.2084
ecf_1R128	-0.4635	-12.4400	0.0373	0.0000	-0.5366	-0.3905
ecf_1R130	-0.6693	-2.4835	0.2695	0.0131	-1.1977	-0.1408
ecf_1R131	-0.2700	-2.3430	0.1152	0.0192	-0.4960	-0.0440
ecf_1R133	-0.3510	-3.6781	0.0954	0.0002	-0.5382	-0.1639
ecf_1R134	0.2994	4.4092	0.0679	0.0000	0.1663	0.4326
ecf_1R135	-0.1155	-2.7084	0.0427	0.0068	-0.1992	-0.0319

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_1R136	-0.1596	-4.3726	0.0365	0.0000	-0.2312	-0.0880
ecf_1R137	-0.6129	-8.8903	0.0689	0.0000	-0.7480	-0.4777

